

Complete quantum information orders

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Abstract

For finite-dimensional channels, unrestricted complete Umegaki relative-entropy order is equivalent to complete symmetric logarithmic derivative order and to simultaneous complete comparison of all operator-convex modular Petz divergences. More generally, one fixed operator-convex generator with finite left endpoint and a nonzero interior representing measure already determines this order. The proof uses a saturated reference-only state pair, second derivatives in the denominator, and realizable rational filters. Every strict violation has a finite-reference witness; explicit bounds are given for conversion from an SLD violation to a relative-entropy violation. Single interior Petz Rényi orders and sandwiched Rényi order two have the same comparison preorder. Petz orders zero and two do not; complete Petz order two instead has an exact finite complex-linear lift characterization. For the interior equivalence class, the argument does not establish a uniform reference bound or a necessary common HP lift. It does not identify numerical contraction coefficients.

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1 Definitions and theorem

All Hilbert spaces are finite dimensional. A channel is completely positive and trace preserving. The channels $\mathcal{P} : A \rightarrow B$ and $\mathcal{Q} : A \rightarrow C$ have the same input, but their output dimensions may differ. Write $\mathcal{P}_R = \mathcal{P} \otimes \text{id}_R$. A reference is always untouched by the channel.

For a positive definite W and Hermitian K , define

$$g_W(K) = \text{Tr } K((L_W + R_W)/2)^{-1} K = \sum_{i,j} \frac{2|K_{ij}|^2}{w_i + w_j}, \quad L_W(X) = WX, \quad R_W(X) = XW. \quad (1)$$

The entries in the sum are in an eigenbasis of W . Complete SLD comparison means

$$\mathcal{P} \succeq_{\text{SLD}}^c \mathcal{Q} \iff g_{\mathcal{P}_R(\rho)}(\mathcal{P}_R(H)) \geq g_{\mathcal{Q}_R(\rho)}(\mathcal{Q}_R(H)) \quad (2)$$

for every finite R , faithful input state ρ , and Hermitian H . One may equivalently require $\text{Tr } H = 0$: writing $H = H_0 + c\rho$, the cross term is $2c \text{Tr } H_0 = 0$ and the additional c^2 term is the same on both sides.

Output matrices are compressed to the fixed support $\text{ran } \mathcal{P}_R(\mathbb{I})$ or $\text{ran } \mathcal{Q}_R(\mathbb{I})$. If $\rho \geq a\mathbb{I}$ with $a > 0$, its output is faithful on this support. Every output of a Hermitian input is supported there, since $-\|H\|\mathbb{I} \leq H \leq \|H\|\mathbb{I}$. Consequently (2) makes sense even for channels whose outputs never have full ambient rank.

For an operator-convex function $f : (0, \infty) \rightarrow \mathbb{R}$, the *modular Petz divergence* is

$$D_f(A\|B) = \sum_{i,j} b_j f(a_i/b_j) |\langle u_i, v_j \rangle|^2, \quad (3)$$

where $A = \sum_i a_i |u_i\rangle\langle u_i|$ and $B = \sum_j b_j |v_j\rangle\langle v_j|$. The scalar perspective is extended at the boundary by its usual lower-semicontinuous values, and the $(0, 0)$ contribution is zero. Infinite values are allowed. On faithful matrices this is equivalently the relative-modular-operator definition. This note concerns this quantization, not maximal or measured f -divergences.

Complete f comparison means

$$\mathcal{P} \succeq_f^c \mathcal{Q} \iff D_f(\mathcal{P}_R(\alpha)\|\mathcal{P}_R(\beta)) \geq D_f(\mathcal{Q}_R(\alpha)\|\mathcal{Q}_R(\beta)) \quad (4)$$

for every finite reference and every pair of input states. *No equality of reference marginals is imposed.* The Umegaki choice is $f(x) = x \log x$, giving

$$D(A\|B) = \text{Tr } A(\log A - \log B)$$

for states with $\text{supp } A \subseteq \text{supp } B$, and $+\infty$ otherwise. Logarithms are natural unless explicitly indicated.

After removing an affine function, the canonical operator-convex representation is

$$f(x) = c_0 \frac{(x-1)^2}{x} + c_\infty (x-1)^2 + \int_{(0,\infty)} \frac{(x-1)^2}{x+s} \mu(ds), \quad c_0, c_\infty \geq 0, \quad (5)$$

where μ is positive and $\int (1+s)^{-1} \mu(ds) < \infty$. Affine terms make the same contribution after either trace-preserving channel, so they do not change an order comparison. If $f(0+)$ is finite, then $c_0 = 0$ and $\int s^{-1} \mu(ds) < \infty$.

Theorem 1.1 (Universality of unrestricted complete order). *For finite-dimensional channels $\mathcal{P}, \mathcal{Q}$ with common input, the following are equivalent:*

1. $\mathcal{P} \succeq_{\text{SLD}}^c \mathcal{Q}$.
2. $\mathcal{P} \succeq_{\text{RE}}^c \mathcal{Q}$.
3. $\mathcal{P} \succeq_f^c \mathcal{Q}$ simultaneously for every operator-convex modular Petz generator, with extended support values.
4. Complete comparison holds for the sandwiched Rényi divergence of order two, defined in (29).

Furthermore, fix any operator-convex f with $f(0+) < \infty$ and nonzero interior measure μ in (5). Complete comparison for this single f is equivalent to the four conditions above.

For every fixed $\alpha \in (0, 2)$, complete comparison for the Petz Rényi divergence

$$D_\alpha^{\text{P}}(A\|B) = \frac{1}{\alpha-1} \log \text{Tr } A^\alpha B^{1-\alpha} \quad (6)$$

with the Umegaki limit at $\alpha = 1$ is likewise equivalent to these conditions. The endpoint assertions at $\alpha = 0, 2$ are false.

The positive integral representation in (5) is the standard operator-convex representation. The arguments below prove all channel-order implications, including the reverse entropy-to-SLD implication. No claim of exhaustive historical priority is made. Complete-order and monotone-metric frameworks appear in [1, 2, 3]; different quantizations of quantum f -divergence must not be substituted for (3).

2 From SLD to every modular Petz divergence

For $s > 0$ put $f_s(x) = (x-1)^2/(x+s)$. If ρ, σ are faithful and $X = \rho - \sigma$, their spectral decompositions give

$$D_{f_s}(\rho\|\sigma) = \langle X, (L_\rho + sR_\sigma)^{-1} X \rangle_{\text{HS}}. \quad (7)$$

Indeed, the (i, j) summand in the mixed eigenbases is $(\rho_i - \sigma_j)^2 |\langle u_i, v_j \rangle|^2 / (\rho_i + s\sigma_j)$.

Append an untouched qubit and use the faithful state and traceless Hermitian tangent

$$\Omega_s = \frac{1}{1+s} \begin{pmatrix} \rho & 0 \\ 0 & s\sigma \end{pmatrix}, \quad H_X = \begin{pmatrix} 0 & X \\ X & 0 \end{pmatrix}. \quad (8)$$

Solving the two off-diagonal Sylvester equations in (1) yields

$$g_{\Omega_s}(H_X) = 4(1+s)D_{f_s}(\rho\|\sigma). \quad (9)$$

The same identity holds after either channel. Complete SLD comparison therefore implies complete f_s comparison for every $s > 0$ and every faithful pair.

The boundary extension is elementary for these kernels. The scalar function

$$h_s(a, b) = \frac{(a - b)^2}{a + sb}, \quad h_s(0, 0) = 0,$$

is continuous on the nonnegative quadrant. The commuting positive superoperators L_A, R_B give $D_{f_s}(A \| B) = \langle \mathbb{I}, h_s(L_A, R_B) \mathbb{I} \rangle_{\text{HS}}$. Hence the divergence is continuous on positive semidefinite matrix pairs in fixed dimension. Regularizing the input states by a common faithful state extends the comparison to arbitrary pairs.

As $s \downarrow 0$, $h_s(a, b)$ increases to the perspective of $(x - 1)^2/x$. As $s \uparrow \infty$, $sh_s(a, b)$ increases to the perspective of $(x - 1)^2$. These limits include infinite values. We therefore obtain both endpoint comparisons. Finally, remove the affine part of f and integrate the nonnegative terms in (5). Tonelli's theorem applies, including at singular matrices. This proves condition 1 implies condition 3 in Theorem 1.1. Condition 3 implies condition 2 immediately.

3 The relative entropy reverse implication

The main construction differentiates the *second* argument at a noncommuting reference pair. Differentiating relative entropy only at equal numerator and denominator gives the BKM metric and does not by itself give SLD.

3.1 A saturated reference pair

Fix an arbitrary original reference and absorb it into both channels. Let ρ be faithful and $H = H^*$ arbitrary. Choose an additional reference with distinguished vector e_0 and leaf coordinates $1, \dots, N$. Set

$$W = \text{diag}(1, \lambda_1, \dots, \lambda_N), \quad \lambda_k > 0, \quad \tau = vv^*, \quad L = e_0 w^* + w e_0^*, \quad Z = \text{Tr } W, \quad (10)$$

where v, w are leaf-supported and $\|v\| = 1$. Define normalized input states

$$\alpha = \rho \otimes \tau, \quad \beta(\varepsilon) = \frac{\rho \otimes W + \varepsilon H \otimes L}{Z}. \quad (11)$$

The denominator has trace one since $\text{Tr } L = 0$, even when $\text{Tr } H \neq 0$. It is positive definite for a sufficiently small interval of both signs of ε . The pure numerator causes no problem because the denominator is faithful.

For either channel Φ , write $A = \Phi(\rho)$ and $K = \Phi(H)$, on its effective output support. At zero,

$$D(A \otimes \tau \| A \otimes W/Z) = D(\tau \| W/Z). \quad (12)$$

This value is independent of the channel. Thus if complete relative-entropy comparison holds, the source-minus-target difference $\Delta(\varepsilon)$ is nonnegative near zero with $\Delta(0) = 0$, and $\Delta''(0) \geq 0$. Conjugation on the reference by $\text{diag}(-1, 1, \dots, 1)$ changes ε to $-\varepsilon$ while leaving the numerator unchanged. Each entropy along the path is even, so its first derivative also vanishes individually.

3.2 The exact denominator Hessian

For positive B and Hermitian Y , differentiation of the logarithm resolvent formula gives

$$-\frac{d^2}{d\varepsilon^2} \log(B + \varepsilon Y) \Big|_0 = 2 \int_0^\infty (B + u)^{-1} Y (B + u)^{-1} Y (B + u)^{-1} du. \quad (13)$$

Normalization by Z in (11) only adds $\log Z$ to the entropy and does not alter this Hessian.

Let a_i be the eigenvalues of A and define

$$F(t) = w^*(W_{\text{leaf}} + t\mathbb{I})^{-1}v. \quad (14)$$

The ordered system-index contribution to the entropy Hessian is

$$2|K_{ij}|^2 a_i \int_0^\infty \frac{|F(u/a_i)/a_i|^2}{a_j + u} du.$$

The two exterior resolvents act on leaf coordinates; the middle resolvent acts on the centre coordinate. Substitution $u = a_i t$ proves

$$\frac{d^2}{d\varepsilon^2} D \left(A \otimes \tau \left\| \frac{A \otimes W + \varepsilon K \otimes L}{Z} \right\|_0 \right) = 2 \sum_{i,j} |K_{ij}|^2 \int_0^\infty \frac{|F(t)|^2}{a_j + a_i t} dt. \quad (15)$$

All terms are finite: F is bounded near zero and is $O(t^{-1})$ at infinity.

Every finite simple-resolvent sum $F(t) = \sum_{k=1}^N c_k/(t + \lambda_k)$ is realizable. Take $v_k = N^{-1/2}$ and $w_k = \sqrt{N} \bar{c}_k$. Coefficients may be large, but are finite; the two-sided interval in (11) can be correspondingly small. No bound on $\|L\|$ is required for the local implication.

3.3 Explicit finite rational filters

For $m \geq 1$ use $N = 2m + 1$ leaves and put

$$\lambda_j = 1 + \frac{j}{m^3} \quad (0 \leq j \leq 2m), \quad F_m(t) = \frac{t^m}{\prod_{j=0}^{2m} (t + \lambda_j)}. \quad (16)$$

These distinct positive poles give the exact partial fractions

$$F_m(t) = \sum_{j=0}^{2m} \frac{c_j}{t + \lambda_j}, \quad c_j = \frac{(-\lambda_j)^m}{\prod_{k \neq j} (\lambda_k - \lambda_j)}. \quad (17)$$

Every coefficient is real and rational. Let $f_m(t) = t^m/(1+t)^{2m+1}$. From $\log(1+x) \leq x$,

$$\exp \left(-\frac{2}{m} - \frac{1}{m^2} \right) \leq \frac{F_m(t)}{f_m(t)} \leq 1 \quad (t > 0). \quad (18)$$

The ratio has its continuous interpretation at zero. The normalized f_m^2 density is beta-prime with parameters $(2m+1, 2m+1)$. Its mean and variance are

$$\mathbb{E}T_m = \frac{2m+1}{2m}, \quad \text{Var } T_m = \frac{(2m+1)(4m+1)}{(2m-1)(2m)^2}. \quad (19)$$

It concentrates at one. Squaring and normalizing (18) shows that the normalized F_m^2 measure has the same limit, even in total-variation distance relative to this baseline sequence. Consequently, with $I_m = \int_0^\infty |F_m(t)|^2 dt$,

$$\frac{1}{I_m} \int_0^\infty \frac{|F_m(t)|^2}{b + at} dt \longrightarrow \frac{1}{a+b} \quad (a, b > 0). \quad (20)$$

The kernel is bounded and continuous; convergence is uniform when (a, b) ranges over a compact subset of $(0, \infty)^2$.

Complete entropy comparison supplies the ordering of (15) for each finite m . Divide by $I_m > 0$ and use (20). The resulting quadratic form is exactly (1). The original reference was arbitrary, so this proves condition 2 implies condition 1 in Theorem 1.1. Each auxiliary reference has dimension $2m + 2$. No infinite-dimensional reference has been used.

If the original entropy hypothesis is stated only for faithful numerators, replace τ by $(1 - \eta)\tau + \eta\mathbb{I}/(N + 1)$. Saturation persists, and the second-argument Hessian is linear and continuous in this numerator. Letting $\eta \downarrow 0$ recovers the calculation.

4 One general finite endpoint generator suffices

This section proves the stronger single-generator assertion. Fix f with finite $f(0+)$ and nonzero interior measure in (5).

4.1 State comparison implies unnormalized comparison

Lemma 4.1 (Reference padding). *Unrestricted complete f comparison on states implies the same comparison on arbitrary positive unnormalized input pairs for which the divergences are defined.*

Proof. For a pair Σ, Ω , choose $\gamma > 0$ such that $\gamma \text{Tr } \Sigma < 1$ and $\gamma \text{Tr } \Omega < 1$. In a new orthogonal reference sector append a fixed input state ρ_0 and use

$$\alpha = \gamma \Sigma \oplus (1 - \gamma \text{Tr } \Sigma) \rho_0, \quad \beta = \gamma \Omega \oplus (1 - \gamma \text{Tr } \Omega) \rho_0.$$

Both are normalized. Petz divergences are jointly homogeneous and additive over direct sums. If $p = 1 - \gamma \text{Tr } \Sigma$ and $q = 1 - \gamma \text{Tr } \Omega$, either channel gives the padding contribution $qf(p/q)$, since its output on ρ_0 has trace one. The comparison difference is therefore exactly γ times the original difference. In a trace-preserving denominator perturbation the same padding weights work throughout the interval. $\square$

One must not normalize only Ω and treat a general f as unchanged. Lemma 4.1 removes that normalization issue.

4.2 Exact resolvent Hessians

Keep the saturated construction, but now use raw matrices

$$\Sigma = A \otimes vv^*, \quad \Omega = A \otimes W, \quad W = \text{diag}(c, \lambda_1, \dots, \lambda_N), \quad c > 0, \quad Y = K \otimes L. \quad (21)$$

The scalar perspective has the exact division

$$tf_s(a/t) = \frac{t}{s} - \frac{a(2s+1)}{s^2} + \frac{a^2(1+s)^2}{s^2(a+st)}. \quad (22)$$

For fixed numerator Σ with positive eigenpairs (a_i, z_i) ,

$$D_f(\Sigma \| T) = \sum_i \langle z_i, T f(a_i T^{-1}) z_i \rangle + f(0) \text{Tr}(\Pi_{\ker \Sigma} T).$$

The zero-eigenvalue term is affine in the denominator. Differentiating the inverse in (22) twice and applying the centre/leaf calculation gives

$$\left. \frac{d^2}{d\varepsilon^2} D_{f_s}(\Sigma \| \Omega + \varepsilon Y) \right|_0 = 2(1+s)^2 |G(s)|^2 \sum_{i,j} \frac{|K_{ij}|^2}{a_i + sca_j}, \quad G(s) = \sum_k \frac{\overline{w_k} v_k}{1 + s\lambda_k}. \quad (23)$$

To check the scale, the inverse in the i th numerator term is $(a_i \mathbb{I} + s\Omega)^{-1}$. Its two exterior leaf factors contribute $a_i^{-2}|G(s)|^2$, canceling the numerator's a_i^2 in (22). The middle centre factor has denominator $a_i + sca_j$.

At $\varepsilon = 0$, $D_f(A \otimes \tau \| A \otimes W) = D_f(\tau \| W)$, independent of the channel. The path is even by the same sign conjugation as before. Thus complete comparison and padding imply an ordered Hessian for the integral of (23).

4.3 Removing the quadratic endpoint and differentiating the integral

For $q(x) = (x - 1)^2$,

$$D_q(\Sigma \| \Omega) = \text{Tr } \Sigma^2 \Omega^{-1} - 2 \text{Tr } \Sigma + \text{Tr } \Omega.$$

Its denominator Hessian vanishes if

$$\sum_k \frac{\overline{w_k} v_k}{\lambda_k} = 0, \quad \text{equivalently} \quad \lim_{s \rightarrow \infty} sG(s) = 0. \quad (24)$$

In fact the entire quadratic divergence is constant on the perturbation: (24) implies $Y\Omega^{-1}$ annihilates the numerator support, so $(\Omega + \varepsilon Y)^{-1}$ agrees there with Ω^{-1} ; also $\text{Tr } Y = 0$.

For completeness, differentiation under the representing integral is justified uniformly. Choose a two-sided interval with $\Omega + \varepsilon Y \geq d\mathbb{I}$ for $d > 0$. The positive-eigenvalue formula and (22) imply

$$\left| \frac{d^2}{d\varepsilon^2} D_{f_s}(\Sigma \| \Omega + \varepsilon Y) \right| \leq 2(1+s)^2 \|Y\|^2 \sum_i \frac{a_i^2}{(a_i + sd)^3}. \quad (25)$$

The right side is $O(1)$ near zero and $O(s^{-1})$ at infinity. It is integrable against μ under the canonical integrability condition. Each first derivative at zero vanishes. Taylor's integral formula and dominated convergence therefore give precisely the integrated Hessian in (23). The finite value of $f(0+)$ also ensures that the singular numerator itself has finite divergence against a faithful denominator.

4.4 Concentrating an arbitrary interior measure

Choose an interior support point $s_0 > 0$ of the nonzero positive measure μ and set $c = 1/s_0$. Consider

$$G_n(s) = \frac{s^n}{(s + s_0)^{2n+2}}, \quad \nu(ds) = \frac{(1+s)^2}{(s + s_0)^4} \mu(ds), \quad h(s) = \frac{s}{(s + s_0)^2}. \quad (26)$$

The measure ν is finite: its weight is bounded near zero and $O(s^{-2})$ at infinity. It has the same interior support as μ . Moreover

$$(1+s)^2 |G_n(s)|^2 \mu(ds) = h(s)^{2n} \nu(ds).$$

The function h has a unique strict maximum at s_0 and tends to zero at both endpoints. Given a neighborhood U of s_0 , choose a smaller neighborhood V so that $\inf_V h > \sup_{U^c} h$. Since $\nu(V) > 0$,

$$\frac{\int_{U^c} h^{2n} d\nu}{\int h^{2n} d\nu} \leq \frac{\nu((0, \infty))}{\nu(V)} \left(\frac{\sup_{U^c} h}{\inf_V h} \right)^{2n} \rightarrow 0.$$

This works for continuous, singular, or atomic representing measures.

The repeated-pole model is replaced by an exactly realizable finite filter:

$$r_j = s_0 \left(1 + \frac{j}{n^3} \right) \quad (0 \leq j \leq 2n+1), \quad \tilde{G}_n(s) = \frac{s^n}{\prod_{j=0}^{2n+1} (s + r_j)}. \quad (27)$$

Uniformly for $s \geq 0$, with the limiting interpretation at zero,

$$e^{-(2/n+3/n^2+1/n^3)} \leq \tilde{G}_n(s)/G_n(s) \leq 1.$$

The normalized weighted measures therefore still concentrate at s_0 . Write the simple partial fractions as $\sum_j d_j/(s+r_j)$, and set $\lambda_j = 1/r_j$, $c_j = d_j/r_j$. Then $\tilde{G}_n(s) = \sum_j c_j/(1+s\lambda_j)$ is realizable by normalized constant v and appropriately scaled w . Since $\tilde{G}_n(s) = O(s^{-n-2})$, its residues satisfy $\sum_j d_j = 0$. Hence $\sum_j c_j/\lambda_j = 0$ exactly, proving (24) without approximation.

Divide the Hessian comparison by

$$Z_n = \int (1+s)^2 |\tilde{G}_n(s)|^2 \mu(ds) > 0.$$

The bounded continuous kernels $(a_i + sca_j)^{-1}$ converge under these normalized measures to $(a_i + a_j)^{-1}$. Formula (23) therefore gives the SLD comparison. There are $2n+2$ leaves, one centre, and at most one padding coordinate. Each reference is finite. The original reference was arbitrary, proving the single-generator assertion of Theorem 1.1.

If comparison was initially restricted to faithful numerators, approximate τ by $(1-\eta)\tau + \eta\mathbb{I}/(N+1)$. At each fixed denominator, continuity of the finite-dimensional spectral formula, using $f(0+) < \infty$, gives the pure-numerator inequality as $\eta \downarrow 0$. Take the denominator derivative only after this limit. Endpoint cancellation need not hold for the intermediate regularized numerator.

5 Interior Petz Renyi orders and BKM order

For $0 < \alpha < 1$, $f_\alpha(x) = -x^\alpha$ is operator convex; for $1 < \alpha < 2$, $f_\alpha(x) = x^\alpha$ is operator convex. Both have finite left endpoint and a nonzero interior representing measure. The logarithm in (6), together with its sign $1/(\alpha-1)$, shows that comparison of the Rényi divergences is exactly comparison of these corresponding f_α divergences. This proves the Rényi assertion from the preceding section.

There is also a direct check of the filtering mechanism. Put $\beta = 1 - \alpha \in (-1, 1) \setminus \{0\}$. In the saturated construction (10), let

$$S_A(\varepsilon) = \text{Tr}[(A^\alpha \otimes \tau)(A \otimes W + \varepsilon K \otimes L)^\beta].$$

Its common value at zero is $S_0 = \text{Tr}(\tau W^\beta) > 0$, and its first derivative is zero. The power resolvent identity is

$$\left. \frac{d^2}{d\varepsilon^2} (B + \varepsilon Y)^\beta \right|_0 = -\frac{2 \sin(\pi\beta)}{\pi} \int_0^\infty u^\beta (B+u)^{-1} Y (B+u)^{-1} Y (B+u)^{-1} du.$$

For positive β it follows from the integral of $u^{\beta-1} B(B+u)^{-1}$; for negative β it follows from the integral of $u^\beta (B+u)^{-1}$. Substitution $u = a_i t$ cancels the system power $a_i^{\alpha+\beta-1} = 1$. Thus

$$(D_\alpha^P)''(0) = \frac{2 \sin(\pi\beta)}{\pi \beta S_0} \sum_{i,j} |K_{ij}|^2 \int_0^\infty \frac{t^\beta |F(t)|^2}{a_j + a_i t} dt. \quad (28)$$

The prefactor is positive throughout the open interval. The filters (16) now have a beta-prime baseline with parameters $2m + \beta + 1$ and $2m + 1 - \beta$. Its mean tends to one and variance to zero, giving the same SLD limit. No uniformity as α approaches an endpoint is claimed.

For comparison, define

$$g_W^{\text{BKM}}(K) = \int_0^\infty \text{Tr}[K(W+u)^{-1} K(W+u)^{-1}] du.$$

The numerator Hessian of $D(W + tK\|W)$ at zero is $g_W^{\text{BKM}}(K)$. Conversely, for faithful states ρ, σ and $X = \rho - \sigma$,

$$D(\rho\|\sigma) = \int_0^1 (1-t) g_{\sigma+tX}^{\text{BKM}}(X) dt.$$

This follows by twice integrating the second derivative of $\text{Tr}(\sigma + tX) \log(\sigma + tX)$. Therefore complete BKM comparison and complete Umegaki comparison are equivalent; arbitrary state pairs follow by the usual common faithful regularization. In particular complete BKM belongs to the same order equivalence of Theorem 1.1.

6 Sandwiched Renyi order two

The sandwiched order-two divergence is

$$\tilde{D}_2(A\|B) = \log \tilde{Q}_2(A\|B), \quad \tilde{Q}_2(A\|B) = \text{Tr } AB^{-1/2}AB^{-1/2}, \quad (29)$$

with the usual support extension. It is different from the Petz order-two divergence $\log \text{Tr } A^2 B^{-1}$.

Proposition 6.1. *Unrestricted complete $\tilde{D}_2$ comparison is equivalent to the orders of Theorem 1.1.*

Proof. Define the geometric quadratic form

$$g_W^{\text{geo}}(H) = \text{Tr } HW^{-1/2}HW^{-1/2} = \sum_{i,j} \frac{|H_{ij}|^2}{\sqrt{w_i w_j}}.$$

For states ρ, σ with σ faithful,

$$\tilde{Q}_2(\rho\|\sigma) = 1 + g_\sigma^{\text{geo}}(\rho - \sigma). \quad (30)$$

This follows by expanding $\rho = \sigma + H$ and using $\text{Tr } H = 0$. Thus complete geometric-metric comparison and complete sandwiched order-two comparison are equivalent, first for faithful states and then by support regularization.

Complete SLD comparison implies comparison of every kernel $\sum_{i,j} |H_{ij}|^2/(w_i + tw_j)$: apply the coherent block test with base $\text{diag}(\rho, t\rho)/(1+t)$ and off-diagonal tangent H . Integrating the resulting inequality against $t^{-1/2}dt/\pi$ gives geometric-metric comparison, because

$$\frac{1}{\pi} \int_0^\infty \frac{t^{-1/2}}{a+tb} dt = \frac{1}{\sqrt{ab}}.$$

This proves the forward implication.

For the reverse, use the normalized saturated pair (11), with raw denominator $B_\varepsilon = A \otimes W + \varepsilon K \otimes L$. Let $V_v x = x \otimes v$ and compress

$$C_\varepsilon = V_v^* B_\varepsilon^{-1/2} V_v, \quad c = v^* W^{-1/2} v > 0.$$

The raw sandwiched overlap is

$$Q(\varepsilon) = \text{Tr } AC_\varepsilon AC_\varepsilon.$$

At zero, $C_0 = cA^{-1/2}$, $Q(0) = c^2$, and $C'_0 = 0$ by the centre/leaf symmetry. Therefore

$$Q''(0) = 2c \text{Tr } A^{3/2} C''_0.$$

The inverse-square-root resolvent formula gives

$$C''_0 = \frac{2}{\pi} \int_0^\infty u^{-1/2} V_v^* (B_0 + u)^{-1} (K \otimes L) (B_0 + u)^{-1} (K \otimes L) (B_0 + u)^{-1} V_v du.$$

The same leaf/centre calculation and substitution $u = a_i t$ now yield

$$Q''(0) = \frac{4c}{\pi} \sum_{i,j} |K_{ij}|^2 \int_0^\infty \frac{t^{-1/2} |F(t)|^2}{a_j + a_i t} dt. \quad (31)$$

Normalizing the denominator multiplies Q by Z , which disappears upon taking the second derivative of its logarithm. Hence the divergence Hessian is the right side of (31) divided by c^2 , with common positive prefactor $4/(\pi c)$.

The zero-order divergence is reference-only, so complete comparison orders these Hessians. The explicit filters (16), normalized with weight $t^{-1/2}$, concentrate at one by the beta-prime argument with $\beta = -1/2$. Dividing by their positive normalizations and taking the limit gives complete SLD comparison. Pure numerators can be obtained from faithful ones by continuity at each fixed faithful denominator before differentiating. $\square$

The proposition is only a statement about the sandwiched order specified here. No equivalence for every sandwiched Rényi order is inferred.

7 Explicit finite relative entropy witnesses

Suppose a faithful input ρ , Hermitian H , and fixed original reference give a strict SLD violation

$$\delta = g_{A_Q}(K_Q) - g_{A_P}(K_P) > 0, \quad A_\Phi = \Phi(\rho), \quad K_\Phi = \Phi(H).$$

Let

$$\mu = \min\{\lambda_{\min} A_P, \lambda_{\min} A_Q\} > 0, \quad T = \text{Tr } K_P^2 + \text{Tr } K_Q^2.$$

Use effective output supports throughout.

Proposition 7.1 (Finite filter and perturbation bounds). *Choose an integer*

$$m \geq \max \left\{ 6, \left(\frac{8eT}{\mu^2 \delta} \right)^2 \right\}. \quad (32)$$

Construct (11) using (16)–(17). Write $\ell = \|L\|$, $\mu_0 = \lambda_{\min} \rho$, $K_0 = H$, and $\mu_\Phi = \lambda_{\min} A_\Phi$. Omit a ratio when its K_r is zero and define

$$\varepsilon_{\text{pos}} = \min_{r \in \{0, P, Q\}} \frac{\mu_r}{2\|K_r\|\ell}, \quad (33)$$

$$M = 16\ell^3 \left[\left(\frac{\|K_P\|}{\mu_P} \right)^3 + \left(\frac{\|K_Q\|}{\mu_Q} \right)^3 \right]. \quad (34)$$

For every

$$0 < \varepsilon \leq \min\{\varepsilon_{\text{pos}}, I_m \delta / (4M)\}, \quad (35)$$

the pair is admissible and its source-minus-target relative-entropy difference is strictly negative. The added reference dimension is $2m + 2$.

Proof. The baseline beta-prime law satisfies

$$\mathbb{E}(T_m - 1)^2 = \frac{2(m+1)}{m(2m-1)} \leq \frac{4}{m}.$$

The normalized squared-filter density is at most $e^{4/m+2/m^2}$ times this baseline density, hence at most e times it for $m \geq 6$. Its mean absolute distance from one is at most $2e/\sqrt{m}$. Because output-state eigenvalues lie in $[\mu, 1]$,

$$\left| 2\mathbb{E} \frac{1}{b+at} - \frac{2}{a+b} \right| \leq \frac{4e}{\mu^2 \sqrt{m}}.$$

The combined normalized Hessian error is at most $4eT/(\mu^2 \sqrt{m}) \leq \delta/2$. Thus $\Delta''(0) \leq -I_m \delta/2$.

The minimum eigenvalue of the raw reference matrix is one, so (33) preserves at least half the unperturbed least eigenvalue of every input and output denominator. For a positive $B + \varepsilon Y$,

$$\left\| \frac{d^3}{d\varepsilon^3} \log(B + \varepsilon Y) \right\| \leq 6\|Y\|^3 \int_0^\infty (\lambda_{\min}(B + \varepsilon Y) + u)^{-4} du = \frac{2\|Y\|^3}{\lambda_{\min}(B + \varepsilon Y)^3}.$$

Here the third derivative contains six identical ordered terms, each with four resolvents. Division of B, Y by Z cancels in this ratio. The numerator has trace one, giving $|\Delta'''| \leq M$ on the interval. Since $\Delta(0) = \Delta'(0) = 0$, Taylor's theorem gives

$$\Delta(\varepsilon) \leq -\frac{I_m \delta}{4} \varepsilon^2 + \frac{M}{6} \varepsilon^3 \leq -\frac{5I_m \delta}{24} \varepsilon^2 < 0.$$

□

Useful explicit bounds are

$$|c_j| \leq 3^m m^{6m}, \quad \ell \leq (2m+1)3^m m^{6m}, \quad I_m \geq 2^{-10m-5}. \quad (36)$$

The residue denominator has absolute value $m^{-6m} j!(2m-j)!$. For the integral bound restrict to $t \in [1/2, 1]$: the numerator squared is at least 2^{-2m} and every denominator factor is at most four. These bounds permit a smaller explicit perturbation without numerically integrating I_m .

The rational filter coefficients have $O(m \log m)$ encoding length per coefficient. The contribution of the filter to $\log(1/\varepsilon)$ is also $O(m \log m)$, in addition to logarithms of conditioning, tangent norms, and the violation gap. This is a bound on witness coefficients and perturbation scale conditional on the supplied input parameters. It is not a complexity theorem for discovering an SLD violation, certifying those parameters, or computing relative entropy from arbitrary real channel data. For a general representing measure the qualitative construction supplies no quantitative neighborhood-mass bound.

8 Endpoint counterexamples and distinctions

8.1 Petz order two is strictly weaker

Let $\mathcal{N}$ be the qubit Pauli channel with probabilities

$$(p_0, p_1, p_2, p_3) = (7/10, 1/10, 1/10, 1/10), \quad K_a = \sqrt{p_a} \sigma_a,$$

and let $\mathcal{E} = \mathcal{N}^c$ be its minimal complement. All four Kraus operators are linearly independent. There is an invertible reshuffling matrix R such that

$$\mathcal{E}(\rho) = R(\rho \otimes \mathbb{I}_2)R^*, \quad \text{Tr}_2(R^*R) = \mathbb{I}_2. \quad (37)$$

Explicitly, with columns labeled by input i and receiver b , take $R_{a,(i,b)} = (K_a)_{bi}$. The first identity follows from the Kraus formula and the second from trace preservation.

For faithful ρ, σ , substitution of (37) and cyclicity of trace give

$$\mathrm{Tr}[\mathcal{E}(\rho)^2 \mathcal{E}(\sigma)^{-1}] = \mathrm{Tr}[(\rho \sigma^{-1} \rho \otimes \mathbb{I}_2) R^* R] = \mathrm{Tr} \rho^2 \sigma^{-1}. \quad (38)$$

The same computation holds with every finite reference, after reordering tensor factors. It extends to supported singular pairs by restriction to the denominator support, and to unsupported pairs because the invertible congruence preserves support containment. Thus $\mathcal{E}$ and the identity channel have equal complete Petz order-two divergences, or equal complete quadratic-generator divergences.

Nevertheless, at $\rho = \mathbb{I}_2/2$, $H = \sigma_z/2$,

$$g_\rho(H) = 1, \quad g_{\mathcal{E}(\rho)}(\mathcal{E}(H)) = \frac{4p_0p_3}{p_0 + p_3} + \frac{4p_1p_2}{p_1 + p_2} = \frac{11}{20} < 1.$$

Complete order-two comparison therefore does not imply even unreferenced SLD comparison. This also explains why the quadratic-only case $\mu = 0$ is excluded from the general theorem. The congruence identity concerns order two and related maximal-divergence phenomena; it cannot be used to infer Umegaki or general modular Petz preservation.

8.2 Petz order zero is also insufficient

The usual order-zero divergence is $D_0(A\|B) = -\log \mathrm{Tr}(\Pi_{\mathrm{supp} A} B)$. Let Δ be the completely depolarizing channel and $\mathcal{Q} = \eta \mathrm{id} + (1 - \eta)\Delta$ for $0 < \eta < 1$. For every reference and input state ρ_{AR} ,

$$\mathrm{supp}((\mathcal{Q} \otimes \mathrm{id}_R)(\rho)) = B \otimes \mathrm{supp} \rho_R = \mathrm{supp}((\Delta \otimes \mathrm{id}_R)(\rho)).$$

The positive depolarizing component supplies the lower support inclusion, while the reference marginal supplies the upper inclusion. Hence both channels give

$$D_0(\Phi_R(\rho)\|\Phi_R(\sigma)) = -\log \mathrm{Tr}(\Pi_{\mathrm{supp} \rho_R} \sigma_R), \quad \Phi \in \{\Delta, \mathcal{Q}\}.$$

Their complete order-zero comparisons coincide. At a faithful unreferenced base, however, the replacer has zero SLD value while $\mathcal{Q}$ has a strictly positive value on a nonzero tangent.

8.3 What the theorem does not identify

The theorem assumes all finite references and unrestricted input pairs. Its saturated pairs have different reference marginals. An equal-reference-marginal-only hypothesis cannot be substituted in the proof.

The result equates channel *preorders*, not numerical contraction coefficients. It does not produce a physical degrading channel. A state-independent HP observable lift with complete-square domination is sufficient, but the proof gives no general necessity theorem for such a lift. It gives no reference dimension depending only on channel dimensions, no gap-independent finite SDP, and no general decision algorithm. The finite-left-endpoint condition is sufficient, not claimed necessary: for example $f(x) = -\log x$ gives reverse relative entropy, which has the same unrestricted pair order by swapping the two input states.

9 Operational consequences with explicit scope

9.1 Tensorization and classical information

Proposition 9.1. *If $\mathcal{P}_i \succeq_{\mathrm{RE}}^c \mathcal{Q}_i$ for $i = 1, \dots, n$, then $\bigotimes_i \mathcal{P}_i \succeq_{\mathrm{RE}}^c \bigotimes_i \mathcal{Q}_i$. In particular the equivalent orders of Theorem 1.1 tensorize.*

Proof. Replace one channel at a time. After applying the other channels, regard all their outputs and the original untouched reference as the reference for the channel being replaced. Its complete relative-entropy comparison applies to the resulting two states. Chaining these n inequalities gives the assertion, including extended values. $\square$

For any finite ensemble $\{p_x, \rho_x\}$, with average $\bar{\rho}$, the Holevo information is

$$\chi(\Phi; \{p_x, \rho_x\}) = \sum_x p_x D(\Phi(\rho_x) \| \Phi(\bar{\rho})).$$

Thus complete relative-entropy order implies the corresponding Holevo comparison for every reference-extended ensemble. Entangled inputs across multiple uses are covered by Proposition 9.1; no product-input restriction is present.

9.2 Complementary dominance and private communication

Suppose $\mathcal{N}^c \succeq_{\text{RE}}^c \mathcal{N}$. For every blocklength and every input ensemble,

$$\chi(B^n) \leq \chi(E^n).$$

This immediately excludes positive private information at every blocklength. A direct finite-code consequence is also available. For a uniform classical message of size M , let Bob's average decoding error be ϵ and let Eve's Holevo leakage be at most η bits. Data processing and Fano's inequality give

$$(1 - \epsilon) \log_2 M \leq \eta + h_2(\epsilon). \quad (39)$$

Indeed $(1 - \epsilon) \log_2 M - h_2(\epsilon) \leq I(M : \widehat{M}) \leq \chi(B^n) \leq \chi(E^n)$. This proves zero unassisted private capacity, and hence zero unassisted quantum capacity. The statement permits arbitrary block encodings but does not include preshared secret resources, feedback protocols, or unrelated adaptive control models.

If another channel $\mathcal{M}$ is antidegradable, its complement physically simulates its receiver; relative-entropy data processing gives $\mathcal{M}^c \succeq_{\text{RE}}^c \mathcal{M}$. Tensorization then yields

$$P(\mathcal{N} \otimes \mathcal{M}) = 0.$$

More generally the same conclusion holds when both channels have complete complementary dominance. These consequences belong to the established complete-order/private-capacity framework [1, 2]; the theorem makes any one of its equivalent comparison tests sufficient.

9.3 A finite sufficient observable certificate

For completeness, suppose there is a complex-linear HP unital map $\mathcal{J} : \mathcal{L}(C) \rightarrow \mathcal{L}(B)$ satisfying

$$\mathcal{P}^* \mathcal{J} = \mathcal{Q}^*,$$

and, for every finite reference and every operator X ,

$$(\mathcal{P}^* \otimes \text{id})[(\mathcal{J} \otimes \text{id})(X)^*(\mathcal{J} \otimes \text{id})(X)] \leq (\mathcal{Q}^* \otimes \text{id})(X^* X). \quad (40)$$

Then $\mathcal{P} \succeq_{\text{SLD}}^c \mathcal{Q}$. To see this, use the exact variational formula

$$g_A(K) = \sup_{X=X^*} \{2 \text{Tr} KX - \text{Tr} AX^2\}.$$

Pull a target observable X back to $\mathcal{J}(X)$; reconstruction preserves the linear term, HP preserves Hermiticity, and (40) bounds its quadratic cost. Taking the supremum gives the metric comparison, at every reference. Theorem 1.1 then supplies all the divergence and operational consequences above. This certificate is sufficient; its general necessity is not proved here.

10 Conclusion

The decisive additional test is a noncommuting auxiliary reference at a saturated state pair. Denominator Hessians expose a tunable positive resolvent kernel. Finite simple-pole filters select the SLD kernel, while signed residues cancel the quadratic endpoint when needed. This yields complete order equivalence from one admissible divergence without assuming a common lift. Every strict SLD failure becomes a finite entropy failure, but the required reference size can depend on the gap and conditioning. These are comparison theorems and finite witness constructions, with the endpoint and computational limitations stated above.

A The finite complex lift characterization of Petz order two

The Petz endpoint has an exact finite characterization, even though it is weaker than the interior orders. The missing condition is Hermiticity preservation.

Let $p = \mathcal{P}^*$ and $q = \mathcal{Q}^*$, and compress the outputs so $\mathcal{P}(\mathbb{I})$ and $\mathcal{Q}(\mathbb{I})$ are faithful. Let the effective target output dimension be b and the source dimension be e .

Theorem A.1. *The following are equivalent:*

1. *Complete all-state-pair Petz Rényi order-two comparison holds from $\mathcal{P}$ to $\mathcal{Q}$.*
2. *That comparison holds for reference dimension $2b^2$ alone.*
3. *There is a complex-linear unital map $J : M_b \rightarrow M_e$, not required to preserve Hermiticity, such that $pJ = q$ and*

$$[q(X_i^* X_j) - p(J(X_i)^* J(X_j))]_{i,j} \geq 0 \quad (41)$$

for every finite family (X_i) .

4. *For one complex vector-space basis $F_1, \dots, F_\ell$ of M_b , $\ell = b^2$, there exist matrices $A_i \in M_e$ satisfying*

$$p(A_i) = q(F_i), \quad [q(F_i^* F_j) - p(A_i^* A_j)]_{i,j=1}^\ell \geq 0. \quad (42)$$

Condition (42) has a finite affine semidefinite formulation. No HP constraints should be added: those would change the characterized order.

A.1 Inverse metrics and complex directions

For faithful W and arbitrary complex X , put

$$r_W(X) = \text{Tr } XW^{-1}X^*.$$

For Hermitian H , expansion gives

$$\text{Tr}(W + tH)^2 W^{-1} = 1 + 2t \text{Tr } H + t^2 r_W(H).$$

Consequently local Petz order-two comparison is equivalent to the Hermitian inverse-metric comparison. Conversely, setting $H = \rho - \sigma$ expresses the complete pair comparison exactly as the same quadratic inequality. Trace-zero restrictions can be removed by subtracting $(\text{Tr } H)W$; both channels acquire the same additional $(\text{Tr } H)^2$.

One additional qubit isolates arbitrary complex directions. For a faithful input ρ already carrying a reference of dimension n , use

$$\Omega_\varepsilon = \frac{1}{1+\varepsilon} \begin{pmatrix} \rho & 0 \\ 0 & \varepsilon\rho \end{pmatrix}, \quad H_X = \begin{pmatrix} 0 & X \\ X^* & 0 \end{pmatrix}.$$

If $W = \Phi_n(\rho)$ and $Y = \Phi_n(X)$, direct multiplication gives

$$\frac{\varepsilon}{1+\varepsilon} r_{\Phi_{2n}(\Omega_\varepsilon)}(\Phi_{2n}(H_X)) = r_W(Y) + \varepsilon r_W(Y^*).$$

Letting $\varepsilon \downarrow 0$, Hermitian comparison at reference $2n$ therefore implies

$$r_{\mathcal{P}_n(\rho)}(\mathcal{P}_n(X)) \geq r_{\mathcal{Q}_n(\rho)}(\mathcal{Q}_n(X)) \quad (43)$$

for arbitrary complex X at reference n .

A.2 A common source score for one target matrix

Fix n and abbreviate the amplified adjoints by p_n, q_n . The variational formula is

$$r_{\mathcal{P}_n(\rho)}(\mathcal{P}_n(X)) = \sup_A \{2 \operatorname{Re} \operatorname{Tr} X^* p_n(A) - \operatorname{Tr} \rho p_n(A^* A)\}. \quad (44)$$

The cost is $A^* A$, not AA^* . This orientation follows by completing the square: the optimizing score before applying a channel is XW^{-1} .

Fix a target score B , and let $x = q_n(B)$, $y = q_n(B^* B)$. The affine set $\mathcal{F} = \{A : p_n(A) = x\}$ is nonempty. Indeed, (43) implies $\ker \mathcal{P}_n \subseteq \ker \mathcal{Q}_n$, and hence $\operatorname{ran} q_n \subseteq \operatorname{ran} p_n$.

For each faithful density ρ , finite-dimensional quadratic duality in (44) gives

$$\inf_{A \in \mathcal{F}} \operatorname{Tr} \rho p_n(A^* A) \leq \operatorname{Tr} \rho q_n(B^* B). \quad (45)$$

For an explicit dual interpretation, the left side equals

$$\sup_X \{2 \operatorname{Re} \operatorname{Tr} X^* x - r_{\mathcal{P}_n(\rho)}(\mathcal{P}_n(X))\}.$$

The quadratic form is strictly positive on source scores because $\mathcal{P}_n(\rho) > 0$, so the constrained minimum is attained and this elementary quadratic conjugacy has no duality gap. Inequality (43) bounds this supremum by its target counterpart, which in turn is at most the target cost of the specified score B .

Set $f(A, \rho) = \operatorname{Tr} \rho[p_n(A^* A) - y]$. For a boundary density, mix it with a fixed faithful density ρ_0 . Since $f(A, \rho_0) \geq -\operatorname{Tr} \rho_0 y$, (45) extends to $\inf_{A \in \mathcal{F}} f(A, \rho) \leq 0$ at every density. The density set is compact and convex; f is convex and continuous in A and affine in ρ . The convex minimax theorem therefore gives

$$\inf_{A \in \mathcal{F}} \max_{\rho \text{ density}} f(A, \rho) = \max_{\rho \text{ density}} \inf_{A \in \mathcal{F}} f(A, \rho) \leq 0.$$

The infimum is attained: the maximum is at least $f(A, \rho_0)$, whose quadratic part is coercive because $\mathcal{P}_n(\rho_0) > 0$. We obtain one A satisfying

$$p_n(A) = q_n(B), \quad p_n(A^* A) \leq q_n(B^* B). \quad (46)$$

A.3 A row selects a linear lift

Take $n = \ell = b^2$ and

$$B = \sum_{i=1}^{\ell} F_i \otimes |1\rangle\langle i|.$$

Apply (46). Multiplying the resulting source score on the left by $E = \mathbb{I}_e \otimes |1\rangle\langle 1|$ preserves reconstruction: $p_\ell(EA)$ is the first row of $p_\ell(A) = q_\ell(B)$, which already has no other rows. Moreover $A^*EA \leq A^*A$. Thus write $EA = \sum_i A_i \otimes |1\rangle\langle i|$. Its square and reconstruction conditions are exactly (42).

Define $J(F_i) = A_i$ and extend complex linearly. Congruence by coefficient matrices extends (42) to every finite family, proving (41). Unitality is automatic. Reconstruction and the square inequality at $\mathbb{I}_b$ imply

$$0 \leq p((J(\mathbb{I}_b) - \mathbb{I}_e)^*(J(\mathbb{I}_b) - \mathbb{I}_e)) \leq 0.$$

Taking the ordinary input trace gives

$$0 = \text{Tr}[\mathcal{P}(\mathbb{I}_a)(J(\mathbb{I}_b) - \mathbb{I}_e)^*(J(\mathbb{I}_b) - \mathbb{I}_e)].$$

Faithfulness of $\mathcal{P}(\mathbb{I}_a)$ forces $J(\mathbb{I}_b) = \mathbb{I}_e$.

Conversely, a lift with (41) satisfies the square inequality at every amplification. Substitute $A = J_n(B)$ in (44); reconstruction preserves the linear term and the Gram inequality decreases the cost. Taking the target supremum proves (43), hence complete Petz order-two comparison. Singular pairs follow by the extended support limit. This proves all equivalences. Only a $2\ell = 2b^2$ reference was needed to obtain the common lift.

The row projection is valid precisely because scores are unconstrained complex matrices. In the HP problem, keeping only one row would generally destroy Hermiticity. Thus this argument does not prove necessity of an HP lift for the interior orders.

A.4 An explicit finite semidefinite formulation

Choose a Stinespring isometry $V : \mathbb{C}^a \rightarrow \mathbb{C}^e \otimes \mathbb{C}^k$ for p , so $p(Z) = V^*(Z \otimes \mathbb{I}_k)V$. Put

$$T = [(A_1 \otimes \mathbb{I}_k)V \cdots (A_\ell \otimes \mathbb{I}_k)V], \quad G_Q = [q(F_i^* F_j)]_{i,j}.$$

Then (42) is equivalent by the Schur complement to the affine LMI

$$\begin{pmatrix} G_Q & T^* \\ T & \mathbb{I}_{ek} \end{pmatrix} \geq 0, \quad p(A_i) = q(F_i) \quad (1 \leq i \leq \ell).$$

The variables are the real and imaginary entries of the A_i . This is a finite SDP characterization of the endpoint order. It is not an unconditional polynomial-time exact bit-complexity result at singular feasibility boundaries.

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